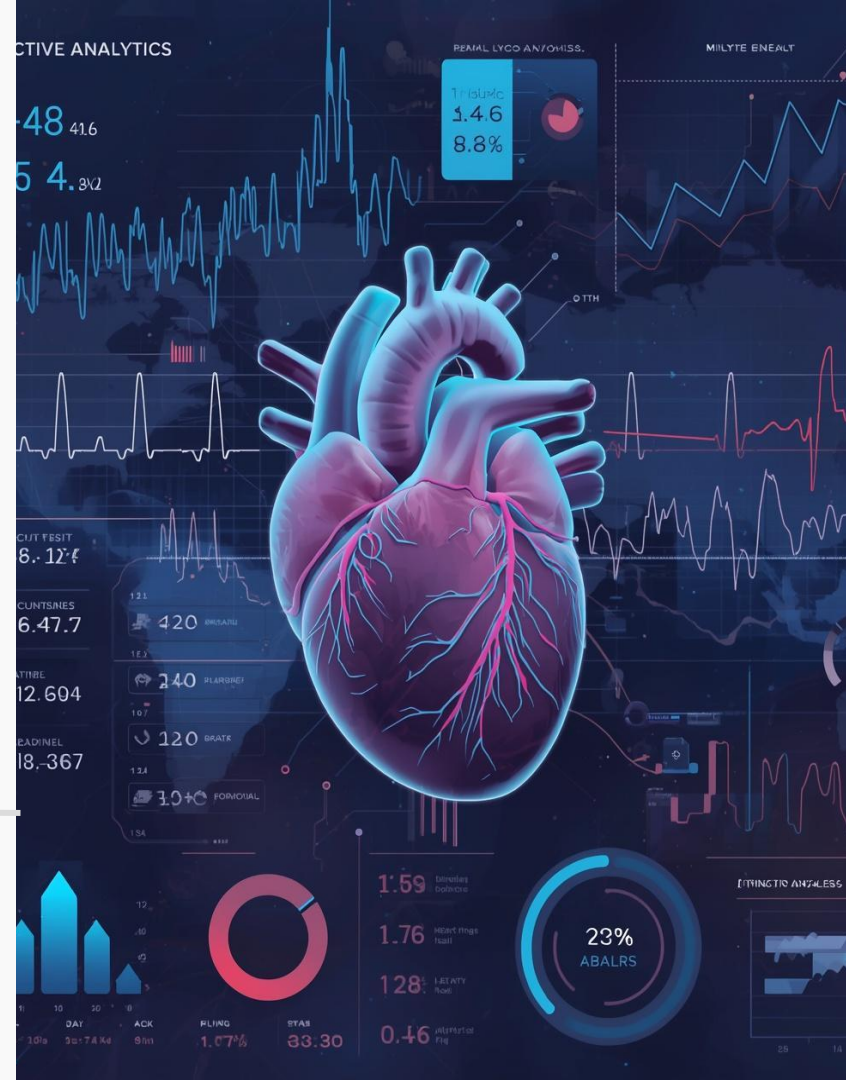


Heart Disease Prediction Using Patient Health Data

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Problem Statement

- Heart disease is a major global health issue
- Early prediction is critical
- Manual diagnosis can be slow and complex
- **Goal:** Predict heart disease using patient data

Objective

Predict presence/absence of heart disease

Use Cleveland dataset (303 records, 13 features)

Apply ML models:

- KNN
- Logistic Regression
- Gradient Boosting

Dataset & Features

13 clinical attributes:

- Age, Cholesterol, Chest Pain (cp)
- Blood Pressure, ECG, Heart Rate (thalach)



Target:

- 0 → No disease
- 1 → Disease

Methodology

Data preprocessing

Feature selection:

- Mutual Information
- Chi-Square

Train-Test Split (80/20)

Scaling using StandardScaler

Models Used

- K-Nearest Neighbors (KNN)
- Logistic Regression
- Gradient Boosting

Compared all models to find best performance

Results

- Logistic Regression → **88.52% (Best)**
- Gradient Boosting → 77%
- KNN → 68%

Logistic Regression selected as final model

Insights & Visualization

Important features:

- Chest Pain (cp)
- Max Heart Rate (thalach)

Used:

- Correlation heatmap
- Decision boundary visualization

Deployment

Built using **Streamlit**

User inputs patient details

Output:

- High Risk
- Low Risk

Real-time prediction system

Conclusion & Future Work

Successfully built prediction system

Helps in early diagnosis

Future improvements:

- Larger dataset
- More features
- Advanced models